

Ajsal Shereef Palattuparambil

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Professional Summary

Ph.D. candidate in Artificial Intelligence with 4+ years of prior applied experience across Machine Learning Engineering, Computer Vision, and Data Science roles. Combines deep theoretical expertise in Reinforcement Learning with hands-on delivery of Generative AI systems, RAG pipelines, autonomous agents, and LLM fine-tuning (LoRA) through to production: containerization, cloud deployment, CI/CD, and observability. Proficient in Python, PyTorch, LangChain, and C++, with a rigorous mathematical foundation and a track record of shipping ML systems end-to-end, not just prototypes.

Technical Skills

- **Core Languages:** Python, C++, R, SQL
- **Data Science & Scientific Computing:** NumPy, Pandas, Scikit-learn, SciPy, Matplotlib, Seaborn, Feature Engineering, Exploratory Data Analysis (EDA), Statistical Inference
- **Generative AI & LLM Engineering:**
 - **Architectures & Techniques:** RAG (Retrieval-Augmented Generation), Agents & Tool Use, LoRA Fine-tuning, Prompt Engineering, Vector Embeddings & Semantic Search, LangGraph State Management.
 - **Frameworks & Tools:** LangChain, LangGraph, Hugging Face, Sentence-Transformers, Gradio, Frontier APIs, Open-Source Models.
 - **Application:** Multi-modal interaction (Text/Image/Audio), Function Calling, Model Evaluation (LLM-as-a-judge).
- **AI/ML Expertise:**
 - **Computer Vision:** Object Detection, Segmentation (Yolact++, SoloV2, UNet), Image Registration, Image Stitching & Feature Matching (ORB)
 - **Deep Learning:** PyTorch, Model Architecture Analysis, Regularization, GANs, VAEs, Diffusion Models
 - **Reinforcement Learning:** RLHF style LLM Fine-tuning, Reward Modeling, Policy Optimisation, Knowledge Transfer, Human-Agent Interaction
 - Practical experience with High Performance Computing
- **MLOps, Deployment & Cloud:**
 - **Cloud & Infrastructure:** Microsoft Azure (Container Apps, Container Registry), Infrastructure-as-Code (Bicep)
 - **CI/CD & Testing:** GitHub Actions, automated testing (pytest, unittest/mock)
 - **Containerization:** Docker (multi-stage builds, container orchestration)
 - **API & Backend:** FastAPI, RESTful API design, React (full-stack prototyping)
 - **Observability:** Azure Application Insights, OpenTelemetry
 - **Edge & Optimisation:** On-device Deployment (Edge AI, TensorFlow Lite), Model Quantization, Performance Benchmarking
- **Developer Tools:** Git, $\LaTeX$
- **Mathematical Foundations:** Probability & Statistics, Linear Algebra, Optimization

Projects

Agentic RAG Assistant via LangGraph

Personal Project

Tech Stack: Python, LangGraph, LangChain, RAG, Vector Databases, LLM Evaluation

- Designed and developed an agentic RAG application utilizing LangGraph for conditional routing, complex state management via custom reducers, and automated tool execution (ToolNode).
- Designed the application architecture with considerations for distributed state persistence, understanding the requirements for transitioning from in-memory state to database-backed checkpoints (SqliteSaver/PostgresSaver) in production.
- Optimized vector document retrieval by engineering strategic chunk overlaps to mitigate context fragmentation and boundary misses, ensuring high-fidelity semantic search.
- Implemented advanced 'LLM-as-a-judge' evaluation pipelines to rigorously test retrieval accuracy (Recall@k, MRR, NDCG) and generation quality (Faithfulness, Answer Relevance), mitigating self-preference bias.

RL Fine-Tuned LLM Agent for Code-Generation Tasks

Personal Project

Tech Stack: Python, PyTorch, LoRA, GRPO, Weights & Biases, Docker, NumPy

- Built an end-to-end reinforcement learning pipeline to fine-tune open-source LLMs via LoRA, training the model to generate working Python code as its output rather than plain text, using GRPO policy-optimization algorithm with KL-regularization against the base model for training stability.
- Designed the reward pipeline: each generated program was executed against a held-out battery of test scenarios and scored on task success, turning code correctness into an RL signal.
- Built a secure sandbox to safely execute untrusted, LLM-generated code (restricted execution, timeout safeguards, isolated state per run), core safety infrastructure for any system where an LLM writes and runs its own code.

- Tuned LoRA and training hyperparameters across multiple open-source model sizes (3B–8B parameters) to fit GPU memory constraints, using Weights & Biases dashboards to monitor training metrics and diagnose noisy reward signals.

AI-Powered Resume-to-Job Matcher

Personal Project

Tech Stack: *Python, FastAPI, Sentence-Transformers, Scikit-learn, Pandas, NumPy, Azure Container Apps, Docker, Bicep, React*

- Architected and deployed an end-to-end ML web application (FastAPI backend, React frontend) that parses resumes and semantically ranks live job listings using Sentence-Transformer embeddings, NumPy vector operations, and Scikit-learn cosine similarity metrics.
- Implemented automated data cleaning, parsing, and text feature extraction pipelines using Pandas and regular expressions to ingest multi-format candidate resumes.
- Containerized the full stack with multi-stage Docker builds and deployed to Azure Container Apps; authored Infrastructure-as-Code (Azure Bicep) covering the registry, compute, and networking layer for fully reproducible, one-command environment provisioning.
- Built and automated a CI/CD pipeline (GitHub Actions) that runs automated test suites (pytest), builds container images, and redeploys to Azure on every commit.
- Instrumented the service for production observability with Azure Application Insights and OpenTelemetry, enabling real-time tracking of latency, throughput, and error rates.

Industry Experience

Machine Learning Engineer

2020 – 2022

Trenser Technology Solutions

Tech Stack: *Python, OpenCV, TensorFlow Lite, PyTorch (Yolact++, SoloV2, UNet), NumPy, Scikit-learn, Scikit-image, C++*

- Led end-to-end development (PoC through production) of an Instax photo-frame digitization pipeline, partnering with an iOS engineer to ship it inside a mobile application now publicly available.
- Re-engineered the frame detection stage by hardening a classical OpenCV/Hough-Transform baseline and introducing a hybrid fallback to object detection and semantic segmentation for noisy, low-contrast cases, lifting detection accuracy by 20 points (71% → 91%) across all frame types.
- Quantized and converted trained models to TensorFlow Lite for real-time, on-device inference on mobile hardware.
- Designed a glare-removal method using multi-angle image stitching (ORB feature matching), exploiting the fact that no single frame region is glare-affected across every capture angle to reconstruct a clean composite output.
- Analyzed, implemented, and benchmarked SOTA computer vision models (e.g., Yolact++, SoloV2, UNet) for medical image segmentation, evaluating performance and efficiency trade-offs to guide deployment decisions.

Junior Data Scientist

2019 – 2020

Lotus Interworks

Tech Stack: *Python, Pandas, NumPy, Scikit-learn, SciPy, Matplotlib, SQL*

- Developed a mathematical evaluation framework and simulation engine using Python (NumPy, SciPy, Pandas) to benchmark call centre agent performance across diverse call volume distributions.
- Built and evaluated machine learning matching algorithms using Scikit-learn to map incoming user query feature vectors to the most relevant service providers.
- Conducted exploratory data analysis (EDA) and data validation routines on tabular transaction records to identify call handling bottlenecks.

Project Associate

2018 – 2019

IIT Madras

Tech Stack: *Python, Pandas, NumPy, Scikit-learn, R (ggplot2, dplyr), Optimization*

- Contributed to the ESPN Cricinfo smart stats project by modelling wicket event probabilities and parameter constraints using Python (Pandas, Scikit-learn, NumPy).
- Performed statistical inference, hypothesis testing, and exploratory data visualization for an environmental monitoring dataset using R (*dplyr*, *ggplot2*).

Education

Ph.D. in Reinforcement Learning

2022 – 2026

Deakin University, Geelong, VIC

Thesis: Towards adapting Reinforcement Learning Agents.

BS-MS, Mathematics as Major and Physics as Minor

2013 – 2018

Indian Institute of Science Education and Research (IISER), Trivandrum, Kerala

Thesis: AKS primality testing and cryptography.

Awards and Scholarships

- **DUPR Scholarship**, Deakin University (2022-2025)
- **INSPIRE Fellowship**, Department of Science and Technology, India (2013-2018)

Research Publications

Conference Proceedings

- A. S. Palattuparambil, T. G. Karimpanal, and S. Rana, "Magik: Mapping to analogous goals via imagination-enabled knowledge transfer," In *Proceedings of the European Conference on Artificial Intelligence (ECAI)*, 2025.
- A. Shereef Palattuparambil, T. Karimpanal George, and S. Rana, "Personalisation via dynamic policy fusion," in *Proceedings of the 12th International Conference on Human-Agent Interaction*, 2024.
- A. Shereef, T. G. Karimpanal, and S. Rana, "PRISM: Policy Reuse via Imagined Semantic Mappings," *NeurIPS 2026*. [Submitted]

Journal Submissions

- A. Shereef, T. G. Karimpanal, and S. Rana, "Dynamic Policy Fusion for User Alignment Without Re-Interaction," *Journal of Ambient Intelligence and Humanized Computing*, 2025. [Under Review]

Teaching and Engagement

Graduate Research Teaching Fellow

Deakin University

- Conducted workshops and marked assessments.

Sessional Academic

Deakin University

- Assisted the lecturer and handled Q&A sessions for course units.

Casual Research Assistant

Deakin University

- Contributed to a cyber security research project.

Maths Tutor

EzyMaths

- Private mathematics home tutoring for grade 9-12.

Technical Committee Member

HARL Workshop

- Peer reviewed papers submitted to the HARL workshop.

Reviewer

AAMAS, NeurIPS

- Reviewed papers submitted to AAMAS 2025 and NeurIPS 2026.

References

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